

Irmak Koseoglu

Machine Learning / AI Engineer

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SUMMARY

Master's student in Big Data and Artificial Intelligence, focused on **Generative AI** and **Machine Learning**. Experienced in building **Retrieval-Augmented Generation (RAG)** systems, training **deep learning** models, and deploying AI applications with **Python** and **PyTorch**. Motivated to build AI systems that work reliably in real products, not only in notebooks.

SKILLS

Languages & Frameworks: **Python, C++, SQL, R**

AI / Machine Learning: **PyTorch, scikit-learn, TensorFlow/Keras, Hugging Face**

LLM / Generative AI: **LangChain, RAG, FAISS, ChromaDB, Prompt Engineering**

Cloud & DevOps: **Docker, Kubernetes, AWS, Microsoft Azure, GitHub Actions (CI/CD)**

Data & Visualization: **Pandas, NumPy, Matplotlib, Jupyter**

Languages: English (fluent), Turkish (native), German (B1)

PROJECTS

PDF Question Answering API — *Retrieval-Augmented Generation Pipeline*

live demo

- Built and deployed a **RAG** pipeline that answers questions from PDF documents: text chunking, **embeddings**, **ChromaDB** retrieval, and answer generation with an LLM.
- Packaged as a **FastAPI** service with automated tests and a **CI/CD** pipeline; runs live in production.

Tech Stack: Python, FastAPI, LangChain, ChromaDB, Docker, GitHub Actions

LLM-Based Data Leakage Detection — *Benchmarking & Evaluation*

- Tested whether an **LLM** can detect data leakage in machine learning pipelines, comparing **prompt engineering** strategies against a rule-based baseline on 96 pipelines.
- Found that **chain-of-thought prompting** gave the best results and was the only method that caught a hard-to-find leakage type.

Tech Stack: Python, OpenAI API, scikit-learn

Plant Disease Classification — *CNN vs. Transfer Learning*

- Trained a **ResNet50** model with **transfer learning** to classify 38 plant diseases from leaf images, clearly outperforming a **CNN** trained from scratch.
- Used **Grad-CAM** to check that the model looks at the correct part of the image before trusting its predictions.

Tech Stack: PyTorch, ResNet50, Grad-CAM, FastAPI, Docker, GitHub Actions

Plant Disease Inference API — *MLOps Deployment*

- Turned the plant disease model into a live **REST API** that returns the top predictions with confidence scores for any app to use.
- Set up a **CI/CD** pipeline that runs tests and builds the **Docker** image on every push, so broken code never reaches deployment.

Tech Stack: FastAPI, Docker, GitHub Actions, pytest, PyTorch

Fake News Detection — *Text Classification*

- Built a model to classify 72,000 news articles as real or fake, using **feature engineering** and **GridSearchCV** to select the best-performing model.

Tech Stack: Python, scikit-learn

EDUCATION

M.Sc. Big Data and Artificial Intelligence, SRH University of Applied Sciences Berlin

October 2025 — March 2027 (expected)

B.Sc. Mathematics Engineering, Istanbul Technical University (ITU)

September 2019 — June 2024

EXPERIENCE

KPMG Turkey

Istanbul, Turkey

Analyst, Digital Finance & Fintech

October 2024 — June 2025

- Supported client projects on data and technology, working with cross-functional teams and reporting results clearly to both technical and non-technical stakeholders.